How Much Targeted Training Does It Take to Top a Multi-Task Leaderboard? Minimum Investment Under Different Aggregation Rules

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Abstract

A leaderboard turns per-task scores into one rank, so a developer can try to lift a chosen model by training on data related to individual sub-datasets. How much such targeted training is needed to put the model first, and how does that depend on the aggregation rule? We define the investment as the minimum raise of the target's task scores, within a per-task cap, that makes it strictly first, and measure it in points, tasks touched and largest single-task jump. We solve it exactly for the mean, median, minimum, Borda and Condorcet-winner rules, prove closed forms for the first three (the median for an odd number of tasks only; OLLB has six), and apply it to every model of the Open LLM Leaderboard (4495 targets, 6 tasks) and of HELM MMLU (78 targets, 57 subjects). With caps taken from observed fine-tuning gains, first place is reachable for only 12% of OLLB targets under the mean (largest feasible rank 901) and 19% of HELM targets (largest feasible rank 16); with a 100-point cap every target can reach it. Among feasible targets the mean is the costliest rule and the median and minimum the cheapest: at OLLB ranks 11–50 the other rules need 0.18–0.50 of the mean's total points. Conclusions depend on the cap, which on HELM is transferred from OLLB, and hold for fixed rivals and score tables only.

1 Introduction

Multi-task benchmarks rank models by aggregating per-task scores into one number, and the resulting rank influences which models are adopted, published and funded. A developer therefore has an incentive to improve a chosen target model specifically on the benchmark's sub-datasets, for instance by training on data related to them. How much such targeted improvement is needed to reach first place, and how that amount depends on the aggregation rule, is the question we study on public leaderboard data.

Our approach is deliberately simple and exact. We take the score table of a public leaderboard as fixed, let the target raise its own scores task by task up to a cap, and compute the *minimum* raise that makes it strictly first under a given aggregation rule. The cap is the central modelling assumption: it stands for how far training on related data can move one score. We therefore report results under a data-grounded cap and an oracle cap, and report infeasibility (no allowed raise makes the target first) as an outcome rather than a missing value.

Contributions. (i) A formulation of the minimum targeted investment with three cost measures (L_1, K, L_∞) and an explicit family of caps (Section 3). (ii) Exact solvers for score rules (mean, median, minimum, power and trimmed means) and rank rules (Borda, Condorcet winner, plurality,

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Copeland; Copeland only on HELM), tested against brute force, with closed forms and proofs for the mean, minimum and odd-n median (Section 3.4, Appendix A). (iii) A study of every model on two public leaderboards, with bootstrap and evaluation-noise intervals (Sections 4–5).

Findings. Under caps grounded in observed fine-tuning gains, first place is out of reach for most models, under every rule: the feasible fraction is 12% (mean) on OLLB and 19% on HELM, and the rule changes how far down the board it is reachable. Where it is reachable, the mean rule costs the most points, and order-statistic and rank rules cost between 18% and 50% of that at OLLB ranks 11–50. Two of three hypotheses fixed beforehand are only partly borne out (Section 5). All conclusions are conditional on the cap and on the other models being fixed; Section 6 lists where they fail.

2 Related work

Benchmarks and leaderboards. MMLU [Hendrycks et al., 2020] aggregates 57 subjects, and HELM [Liang et al., 2022] releases per-scenario scores and raw model outputs for many models; BIG-Bench Hard [Suzgun et al., 2022] selects hard tasks from BIG-Bench. These provide the multi-task score matrices that we analyse.

Fragility of rankings. Dehghani et al. [2021] show that the relative order of methods can change with the choice of benchmark tasks (the "benchmark lottery"). Singh et al. [2025] document how private testing and unequal data access distort the Chatbot Arena leaderboard. Both concern what is measured and disclosed; we instead hold the leaderboard fixed and ask what a single developer must change for one model to win.

Aggregation and social choice. Lanctot et al. [2023] treat each task as a voter and use voting theory to build cross-task evaluation; the Handbook of Computational Social Choice [Brandt et al., 2016] covers the rules (Borda, Condorcet-consistent rules, plurality) we reuse. Our rules are the same families, but our question is the cost of changing the winner, not the choice of rule.

Training on the test task and contamination. Zhou et al. [2023] show that benchmark leakage into training can raise scores sharply, and Yang et al. [2023] show that rephrased test samples evade common decontamination. Chen et al. [2026] analyse, in a game-theoretic model, the incentives that benchmarks create for strategic post-training. Our analysis is empirical and combinatorial: given a fixed score table, we compute the exact minimum targeted improvement per rule, under explicit assumptions about how much any one task can be improved.

Positioning. Closest in question is Chen et al. [2026], who model developers choosing how much to spend on benchmark-specific improvement of a single benchmark score, prove non-existence of equilibria for current protocols and a unique equilibrium for tune-before-test. That analysis concerns one score per model; the choice of aggregation rule over tasks does not enter it. We take the opposite stance: no game and no equilibrium, a fixed table, and the exact cost for one developer under each of several aggregation rules. Colombo et al. [2022] propose a social-choice-based aggregation for NLP benchmarks and show it changes conclusions relative to the mean; we do not propose a rule but measure how costly it is to move a model to first place under each. The computational-social-choice literature studies the analogous *bribery* problem, the cost of changing votes so that a chosen candidate wins under a voting rule [Faliszewski et al., 2009, Brandt et al., 2016]. Our setting differs in that the manipulator changes its own scores on tasks (the "voters") and pays in score points under per-task caps, which gives closed forms for score rules and integer programs for rank rules.

3 Problem setup

3.1 Leaderboard, target and investment

A leaderboard is a score matrix $X \in [0, 100]^{M \times n}$ of M models on n tasks. Fix a target model i with score vector $x = X_{i\cdot}$; all other rows stay fixed. In targeted training the developer can raise the target's score on task t to any $y_t \in [x_t, c_t]$, where c_t is a *cap* (Section 3.2). An aggregation rule f

maps the score vectors to a ranking; the target is *first* if it is strictly above every other model under f (by a margin $\varepsilon = 10^{-3}$ points for score rules). We report the investment in three ways:

$$L_1 = \sum_t (y_t - x_t), \qquad K = \#\{t : y_t > x_t\}, \qquad L_\infty = \max_t (y_t - x_t),$$

the total points, the number of tasks touched, and the largest single-task jump. The minimum of each is taken separately; L_∞ is also the cost of the uniform "general improvement" that raises every task by the same amount. For comparison across rules and datasets we also report K/n and L_1/n.

3.2 Caps: how much can one task be improved?

Without a bound on y_t the question is trivial (put every task at 100). The cap c_t is therefore the central modelling assumption, and we never collapse it to a single choice. Two caps appear in the main text: `ceil`, $c_t = 100$, an oracle upper bound on what can be done and so a lower bound on any required investment; and `emp99`, $c_t = \min(100, x_t + g_t^{99})$ where g_t^{99} is the 99th percentile of the observed gains (fine-tune minus its declared base model) on task t in the Open LLM Leaderboard. `emp99` is the only data-grounded cap. It inherits a selection bias, since only published fine-tunes appear, and it measures gains attained by the community, not the gains a determined developer could attain. Further caps (best existing score $+0$ or $+2$, and at most $+5/+10/+20$ points per task) are reported in the appendix. When no $y \le c$ makes the target first, the case is *infeasible*; infeasibility is reported as a rate and never coded as a large investment, and all investment statistics are conditional on feasibility.

3.3 Aggregation rules

Score rules apply a function f to the score vector of each model: the *mean* (the rule of both leaderboards we use), the *median*, and the *minimum* (worst-task score). Rank rules use only the order of models on each task: *Borda* (sum over tasks of the number of models beaten, ties counting one half; equivalent to mean rank) and the *Condorcet-winner* requirement (the target beats every rival on a strict majority of tasks, ties counting one half). Being a Condorcet winner is sufficient for being first under any Condorcet-consistent rule such as Copeland or Schulze, so its investment upper-bounds theirs. Geometric, harmonic and power means, the trimmed mean and plurality are in the appendix.

3.4 Exact minimum-investment solvers

For a score rule let $\tau = \max_{j \neq i} f(X_{j\cdot})$ be the level of the best rival; the target is first iff $f(y) \ge \tau + \varepsilon$. For the mean, the minimum and the median the minimum investment has a closed form (Propositions 1–3, proved in Appendix A); for power means and the trimmed mean we solve a convex program or enumerate exactly. For rank rules a raise of task t to level $s + \varepsilon$ passes exactly the models with score at most s on that task, so only these finitely many levels need be considered, and we solve the resulting integer program exactly with HiGHS, adding rival constraints lazily. Every solver is checked against brute-force search on small random instances (unit tests in the supplementary code). For each target we report L_1 and K, each minimised separately, and L_∞ for the uniform baseline.

Proposition 1 (Mean)**.** *Let f be the mean and $h_t = c_t - x_t$. The target can become first iff $\sum_t c_t > n\tau$ (up to ε). The minimum L_1 is $n(\tau - f(x))$ regardless of how it is spread, the minimum K is the least k for which the k largest h_t sum to at least this amount, and the minimum L_∞ is $\tau - f(x)$ (raise all tasks equally) when no cap binds.*

Proposition 2 (Minimum)**.** *Let $f = \min_t y_t$. The target can become first iff $c_t > \tau$ for all t with $x_t \le \tau$. Then $L_1 = \sum_t (\tau - x_t)^+$ and $K = \#\{t : x_t \le \tau\}$: every deficient task must be raised.*

Proposition 3 (Median, n odd)**.** *Let $m = (n+1)/2$, $d_t = (\tau - x_t)^+$ for tasks with $c_t > \tau$ and $d_t = \infty$ otherwise. The target is first iff at least m tasks end above τ. The minimum L_1 is the sum of the $(m - \#\{t : x_t > \tau\})^+$ smallest finite d_t among tasks with $x_t \le \tau$, and K equals that count; the target can be made first iff enough finite d_t exist.*

(All statements are for $\varepsilon \to 0$; the code uses $\varepsilon = 10^{-3}$ points. For even n the median is the average of the two middle order statistics and we only have the computed solution.) We verify Propositions 1–3 numerically against the solvers (Section 4); no theorem is claimed for the rank rules.

4 Experimental setup

Data. We use two public leaderboards, each at a fixed, recorded version (checksums in the supplementary reproduction notes). *HELM MMLU*, release v1.13.0 (summary dated 2025-01-10): the exact-match score of 79 models on each of the 57 MMLU subjects (79×57, no missing cell). *Open LLM Leaderboard* (OLLB) `contents`, dataset commit `9c09a7c` (2025-03-20): 4576 rows, of which 4497 have distinct model names after we keep the best-scoring variant of each duplicated name, and 4496 after removing one flagged model; each model has six normalised task scores (IFEval, BBH, MATH Lvl 5, GPQA, MUSR, MMLU-PRO), so $n = 6$. Merged models are kept in the main analysis and dropped in a robustness slice. The mean we use equals the sites' own aggregate (OLLB `Average`, HELM "MMLU All Subjects") to within 6×10^{-14} points on every model. Evaluation noise (Section 5) uses binomial standard errors from item counts: HELM reports them per subject, whereas for OLLB the counts come from a single results file and are assumed equal for all models. Neither source declares a licence for the score data; we redistribute only the derived score tables.

Targets, caps and what is reported. Every model other than the current leader is a target in turn, with all others fixed (4495 targets on OLLB, 78 on HELM). On OLLB the primary cap is `emp99` (per-task 99th percentile of the gain of a fine-tune over its declared base model, estimated from 540 such pairs in the same leaderboard). HELM has no base/fine-tune pairs, so its primary cap, `emp99T`, applies the OLLB MMLU-PRO 99th-percentile gain to all 57 subjects; this is a transfer assumption, not a measurement on HELM. The oracle cap `ceil` is reported beside it. All investment statistics are medians over feasible targets, shown together with the feasible fraction. OLLB rank rules are exact integer programs, which at $M = 4496$ models we run only for the 50 best targets under the mean; we apply a 120 s limit per instance and report time-outs as censored, never as infeasible.

5 Results

All statements below are about targets that can be made first within the cap; the feasible fraction is shown beside every cost, and a rule is called cheaper than another only on targets that are feasible under both.

The closed forms are exact (Q1). Propositions 1–3 agree with the exact solvers on every target of both datasets and every cap: the largest deviation in L_1 is below 2×10^{-12} points, the deviation in K is zero, and no feasibility verdict differs (Appendix Table 3). The OLLB median is solver-only because $n = 6$ is even.

Feasibility depends on rank and on the rule (Q3). Under the data-grounded caps, targeted training cannot make most models first (Figure 1, left). On OLLB every target is feasible under the mean up to rank 394 and under the median up to rank 90 (Table 1), and the feasible fraction is near zero beyond rank about one thousand; on HELM every target at ranks 2–10 is feasible under the mean, median, Borda and Condorcet rules, and the fraction falls steeply afterwards (Table 2: for ranks 11–50 only 6 of 40 targets are feasible under the mean and 17 of 40 under the median). The minimum rule is the most restrictive on both datasets, since every task below the best rival's weakest score must be raised and any one cap can block it (Proposition 2). The cap, not the rule, decides feasibility far from the top: with the oracle cap `ceil` every target is feasible (Appendix Figure 4).

Table 1 gives the frontier. On OLLB the fraction of targets that can reach first place under `emp99` is 12% under the mean, 10% under the median and 6% under the minimum; every target is feasible only up to rank 394 (mean), 90 (median) and 5 (minimum), and the largest feasible rank is 901, 1207 and 2225. On HELM under `emp99T` the fractions are 19%, 33% and 12%, every target is feasible up to rank 16, 19 and 6, and the largest feasible rank is 16, 29 and 20; the Condorcet requirement reaches rank 26 and Borda 17. Copeland, solved exactly on HELM only, gives the same feasibility verdict as the Condorcet requirement for every target under `emp99T` (25 of 78 targets) and `ceil` (all 78), with L_1 equal to within 10^{-3} points (`paper/aux_numbers.py`), so the upper bound is tight there. On HELM the median is thus more exposed to targeted training than the mean on every column; on OLLB the picture is mixed: the mean is feasible for slightly more targets (12% against 10%) and over a longer all-feasible prefix (394 against 90), while the median has the larger, isolated, largest feasible rank (1207 against 901). The minimum is feasible rarely but, when feasible, cheaply. The largest

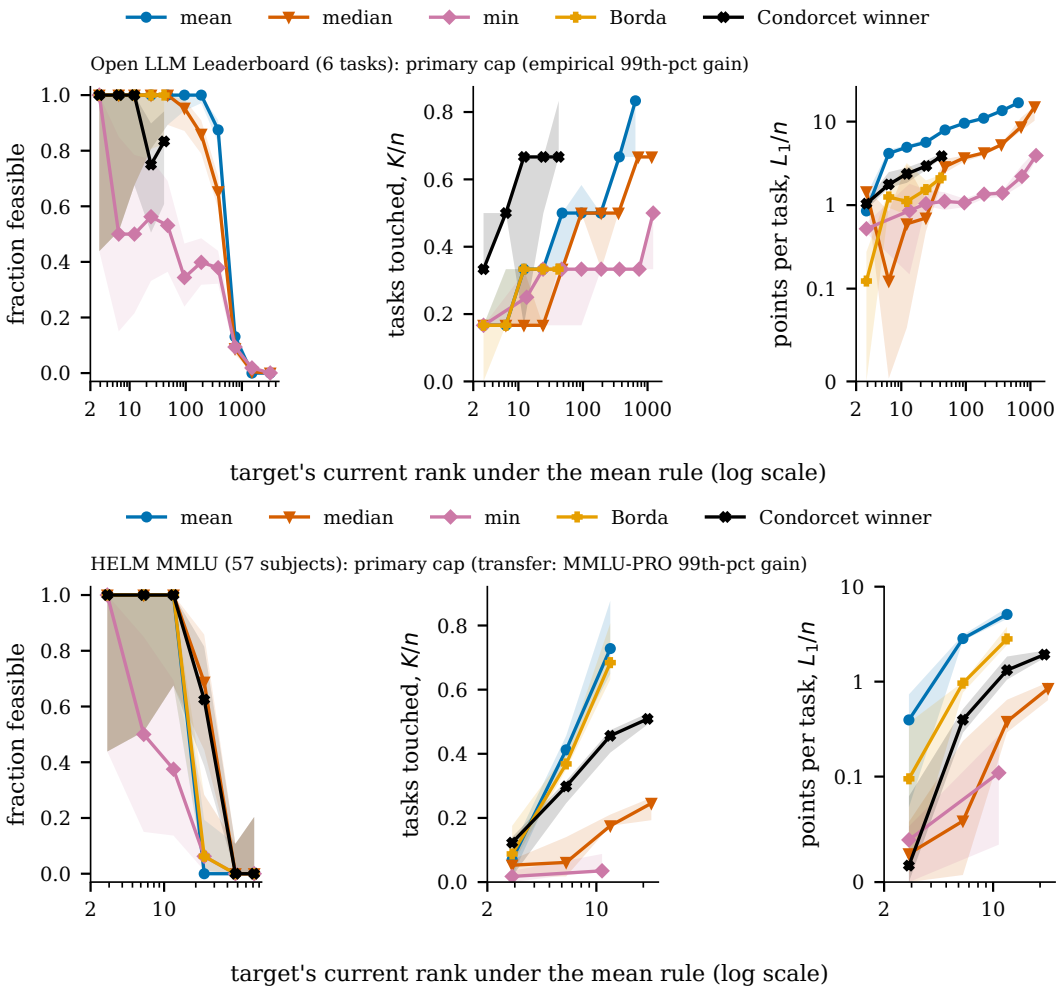


Figure 1: Minimum targeted investment to become first versus the target's current rank under the mean, primary caps. Left: fraction of targets that can reach first at all. Middle and right: tasks touched (K/n) and total points per task (L_1/n), *among feasible targets*. Top: OLLB, $n = 6$, cap `emp99` (4495 targets; Borda and Condorcet only for the 50 best). Bottom: HELM MMLU, $n = 57$, cap `emp99T` (78 targets; the cap is transferred from OLLB). Points are bin medians (bins double from rank 2; bins with fewer than 3 targets omitted), bands are 95% bootstrap intervals over targets. Generated by `figures/src/fig_rank_rules.py`.

feasible rank can be an isolated case (OLLB, minimum), which is why the column "all feasible up to" is shown.

Cost, conditional on feasibility, differs strongly by rule (Q2). Among feasible targets, the mean rule is the most expensive on both datasets and the minimum and median rules the cheapest (Figure 1, right; Table 2). On OLLB at ranks 11–50 the median feasible target needs 37.8 points in total under the mean, 9.1 under the median, 6.9 under the minimum, 11.7 under Borda and 20.5 under the Condorcet-winner requirement; relative to the mean on jointly feasible targets, the ratios are 0.22, 0.18, 0.27 and 0.50. On HELM at ranks 11–50 the corresponding totals are 321, 38, 10, 188 and 105 points, but they are medians over different feasible sets (6, 17, 3, 7 and 16 of 40 targets for the mean, median, minimum, Borda and Condorcet rules), so only the ratios on jointly feasible targets compare rules: 0.09, 0.04, 0.55 and 0.28 for the median, minimum, Borda and Condorcet rules, each on the 6 targets the mean can also reach (or fewer). The reason is structural: the mean must close the whole summed gap (n times the gap, Proposition 1), whereas order-statistic rules pay only for the cheapest majority of tasks or the weakest tasks, and rank rules pay for passing rivals task by task. Pooled over

Table 1: Feasibility frontier under the primary cap (`emp99` on OLLB, `emp99T` on HELM) and the oracle cap (`ceil`). *share*: fraction of targets for which first place is reachable (Wilson 95% intervals); r^*: largest rank under the mean among feasible targets (may be an isolated case); *all feas. up to*: largest r such that every target of rank $\leq r$ is feasible; *first bin $<$ 1/2*: first rank bin (bins double from rank 2) in which fewer than half are feasible. †: the 50 best targets under the mean only, not comparable with the score rules' fractions. Generated by `figures/src/tables.py`.

cap	rule	targets	feasible	share [95% CI]	r^* [95% CI]	all feas. up to [95% CI]	first bin $<$ 1/2
primary (`emp99`)	mean	4495	546	0.12 [0.11, 0.13]	901 [879, 901]	394 [394, 430]	513–1024
primary (`emp99`)	median	4495	451	0.10 [0.09, 0.11]	1207 [1145, 1207]	90 [90, 133]	513–1024
primary (`emp99`)	min	4495	273	0.06 [0.05, 0.07]	2225 [1360, 2225]	5 [5, 9]	65–128
primary (`emp99`)	Borda†	49	49	1.00 [0.93, 1.00]	50 [47, 50]	50 [47, 50]	none
primary (`emp99`)	Condorcet†	49	42	0.86 [0.73, 0.93]	50 [47, 50]	18 [18, 26]	none
oracle (`ceil`)	mean	4495	4495	1.00 [1.00, 1.00]	4496 [4493, 4496]	4496 [4493, 4496]	none
oracle (`ceil`)	median	4495	4495	1.00 [1.00, 1.00]	4496 [4493, 4496]	4496 [4493, 4496]	none
oracle (`ceil`)	min	4495	4495	1.00 [1.00, 1.00]	4496 [4493, 4496]	4496 [4493, 4496]	none
oracle (`ceil`)	Borda†	49	49	1.00 [0.93, 1.00]	50 [47, 50]	50 [47, 50]	none
oracle (`ceil`)	Condorcet†	49	49	1.00 [0.93, 1.00]	50 [47, 50]	50 [47, 50]	none

cap	rule	targets	feasible	share [95% CI]	r^* [95% CI]	all feas. up to [95% CI]	first bin $<$ 1/2
primary (`emp99T`)	mean	78	15	0.19 [0.12, 0.29]	16 [13, 16]	16 [16, 19]	17–32
primary (`emp99T`)	median	78	26	0.33 [0.24, 0.44]	29 [25, 29]	19 [19, 30]	33–64
primary (`emp99T`)	min	78	9	0.12 [0.06, 0.20]	20 [9, 20]	6 [6, 11]	9–16
primary (`emp99T`)	Borda	78	16	0.21 [0.13, 0.31]	17 [14, 17]	17 [17, 20]	17–32
primary (`emp99T`)	Condorcet	78	25	0.32 [0.23, 0.43]	26 [23, 26]	26 [26, 30]	33–64
oracle (`ceil`)	mean	78	78	1.00 [0.95, 1.00]	79 [76, 79]	79 [76, 79]	none
oracle (`ceil`)	median	78	78	1.00 [0.95, 1.00]	79 [76, 79]	79 [76, 79]	none
oracle (`ceil`)	min	78	78	1.00 [0.95, 1.00]	79 [76, 79]	79 [75, 79]	none
oracle (`ceil`)	Borda	78	78	1.00 [0.95, 1.00]	79 [76, 79]	79 [76, 79]	none
oracle (`ceil`)	Condorcet	78	78	1.00 [0.95, 1.00]	79 [76, 79]	79 [76, 79]	none

Table 2: Median investment to become first among feasible targets, by rank stratum and rule, primary caps: OLLB (`emp99`, top) and HELM (`emp99T`, bottom). "feas." = feasible targets / targets in the stratum; K, L_1 with 95% bootstrap intervals over targets; the last column is the median ratio of L_1 to the mean rule over targets feasible under both. OLLB Borda and Condorcet use the 50 best targets only. Generated by `figures/src/tables.py`.

	rank 2–10				rank 11–50			
	feas.	K	L_1	L_1/L_1^{mean}	feas.	K	L_1	L_1/L_1^{mean}
mean	9/9	1.0 [1.0, 2.0]	24.6 [5.1, 27.7]	–	40/40	2.0 [2.0, 2.0]	37.8 [33.9, 41.0]	–
median	9/9	1.0 [1.0, 1.0]	0.8 [0.0, 8.4]	0.03 [0.00, 1.70] (9)	40/40	1.0 [1.0, 2.0]	9.1 [4.4, 14.6]	0.22 [0.13, 0.40] (40)
min	5/9	1.0 [1.0, 2.0]	3.7 [2.8, 14.6]	0.61 [0.23, 0.77] (5)	24/40	2.0 [1.5, 2.0]	6.9 [5.9, 8.4]	0.18 [0.14, 0.19] (24)
Borda	9/9	1.0 [1.0, 2.0]	5.6 [0.7, 6.2]	0.20 [0.09, 0.35] (9)	40/40	2.0 [2.0, 2.0]	11.7 [9.5, 12.6]	0.27 [0.25, 0.31] (40)
Condorcet	9/9	3.0 [2.0, 4.0]	10.0 [6.7, 16.7]	0.59 [0.41, 1.04] (9)	33/40	4.0 [4.0, 4.0]	20.5 [18.0, 22.4]	0.50 [0.47, 0.55] (33)

	rank 2–10				rank 11–50			
	feas.	K	L_1	L_1/L_1^{mean}	feas.	K	L_1	L_1/L_1^{mean}
mean	9/9	23.0 [4.0, 32.0]	156.3 [22.5, 222.4]	–	6/40	46.0 [40.0, 51.0]	321.2 [278.2, 351.3]	–
median	9/9	4.0 [1.0, 10.0]	3.3 [0.4, 16.6]	0.04 [0.00, 0.09] (9)	17/40	12.0 [10.0, 15.0]	37.5 [31.7, 53.6]	0.09 [0.07, 0.11] (6)
min	6/9	1.5 [0.5, 2.5]	2.3 [1.0, 5.3]	0.03 [0.00, 0.09] (6)	3/40	3.0 [2.0, 5.0]	10.4 [6.2, 15.5]	0.04 [0.02, 0.06] (2)
Borda	9/9	20.0 [5.0, 30.0]	54.0 [5.6, 100.5]	0.36 [0.28, 0.94] (9)	7/40	45.0 [38.0, 48.0]	187.8 [143.1, 230.8]	0.55 [0.51, 0.64] (6)
Condorcet	9/9	16.0 [7.0, 22.0]	20.7 [0.9, 57.6]	0.16 [0.02, 0.22] (9)	16/40	28.0 [27.0, 29.0]	104.6 [90.4, 113.9]	0.28 [0.21, 0.31] (6)

all ranks and on targets feasible under both rules, Appendix Table 5 gives the median L_1 ratio to the mean: on OLLB 0.39 for the median (435 of 561 targets), 0.11 for the minimum (237 of 581), 0.27 for Borda and 0.52 for the Condorcet requirement (the last two on the 50 best targets only, so not on the same targets as the first two); on HELM 0.07, 0.05, 0.51 and 0.21 (15 of 26, 7 of 16, 15 of 16, 15 of 25 targets). Pooled, every rule is cheaper than the mean on jointly feasible targets (in the top strata the intervals are wide and can reach above 1, e.g. the OLLB median at ranks 2–5, 1.38 with 4 targets), but the order among the other rules depends on the dataset and on the rank stratum: on OLLB Borda is cheaper than the median pooled (0.27 against 0.39) and in ranks 31–100 (0.27 against 0.37) though not in ranks 11–50 (0.27 against 0.22), and the Condorcet requirement costs more than Borda on OLLB (0.52 against 0.27 pooled) and less on HELM (0.21 against 0.51). Only the cost relative to the mean is established; we do not claim a total order of the other rules. Table 2 shows intervals, which are wide at ranks 2–10 ($n = 9$ targets).

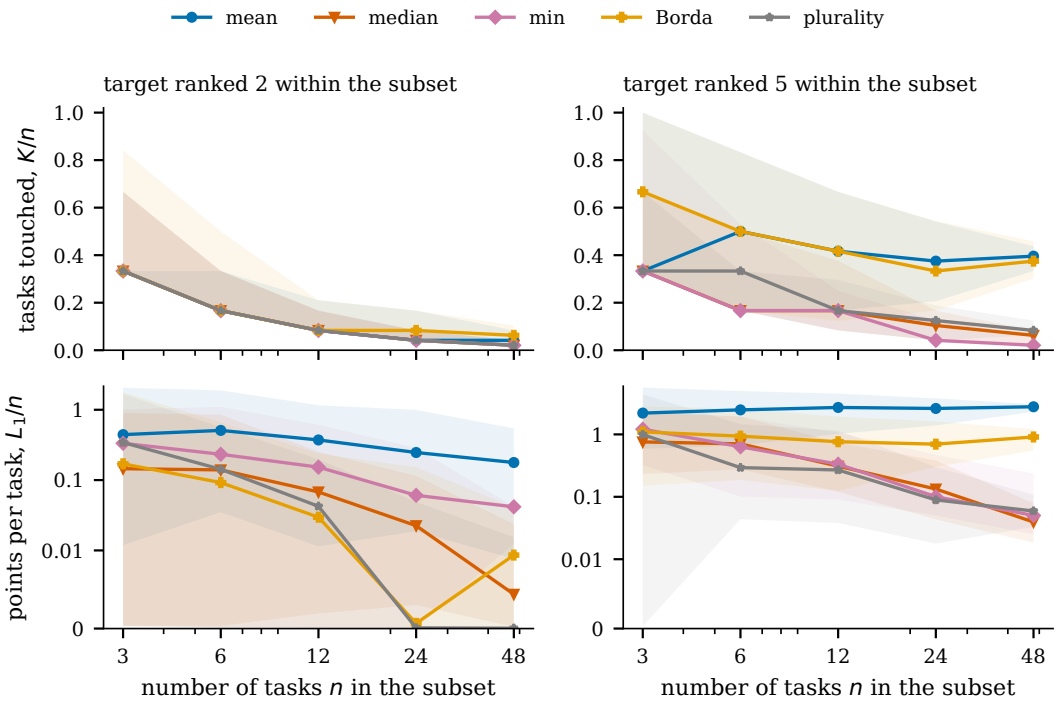


Figure 2: HELM: investment versus the number of tasks n in a random subset of subjects, for the target ranked second (left) or fifth (right) within the subset under the mean; bands span the 2.5–97.5 percentiles over 60 subset draws, among feasible draws, cap `emp99T`. Generated by `figures/src/fig_scaling.py`.

Which of our predictions held. The plan fixed three hypotheses before the full results (H1–H3, Appendix D). H2 (cost rises with rank, feasibility falls as caps tighten) holds on both datasets. H1 (order-statistic and majority rules need *more* tasks but possibly fewer points than the mean) holds only in part: they need fewer points, but not more tasks, except that the Condorcet requirement touches more tasks than the mean on OLLB at $n = 6$ ($K = 4$ against 2 at ranks 11–50); on HELM the median needs fewer tasks than the mean ($K = 12$ against 46). H3 is examined next.

Number of tasks (Q4). Figure 2 draws random subsets of $n \in \{3, 6, 12, 24, 48\}$ HELM subjects and takes the model ranked second or fifth within each subset as the target. For the rank-2 target K/n falls with n under every rule, and for the rank-5 target the mean and Borda rules keep touching about 0.4 of the tasks while the median, minimum and plurality fall. For the rank-5 target the mean rule's L_1/n stays roughly constant in n while that of the median, minimum and plurality falls; for the rank-2 target all rules fall. H3 (that K/n of the median and majority rules does not shrink) is therefore not supported for the median. The bands are wide (60 subset draws), so we make no claim about differences between neighbouring n. OLLB, with $n = 6$, is a single small-n point and supports no scaling claim.

Robustness (Q5). Appendix Figure 3 resamples evaluation noise (binomial standard errors from item counts: 300 draws on OLLB closed-form rules, 60 on OLLB ordinal rules, 200 on HELM, all rules) and tasks (bootstrap over tasks with the same caps; 300 OLLB draws, 60 for ordinal rules, 200 HELM). Where the observed target is feasible, its L_1 lies inside the interval of resampled values; it is often below the draw median, by a few percent for the mean and by up to a factor of about three for the median rule on OLLB at ranks 5–10, so that a single observed investment can understate the typical one under noise. Intervals are narrow for the mean on OLLB beyond rank 10 and wide for the median, the minimum and the top-ranked targets, whose observed gap is small. Feasibility near the frontier is noise-sensitive: a target that is infeasible on the observed data can be feasible in some resampled leaderboards, and the minimum rule is infeasible in a large share of draws on both datasets, while targets far from the frontier (HELM ranks 20 and 40) are infeasible in nearly all draws. A target

whose gap to first place is within the evaluation noise therefore cannot be said to need any particular investment: a win by the margin ε would not survive resampling.

6 Discussion and limitations

What the results say. Under caps taken from observed fine-tuning gains, a developer who can lift only the target's own scores, and whose rivals stand still, tops these leaderboards only from near the top, and how near depends on the rule: on HELM the median reaches further down than the mean, on OLLB the two are close and the evidence is mixed, and the minimum is rarely feasible. Where it is feasible the mean rule demands the most points, because it must close n times the gap (Proposition 1) whatever the spread, while the median and minimum pay only for a majority of tasks or for the weak tasks. Among the score rules the mean is thus the most expensive to manipulate in points; in the rank that can be reached it is the hardest on HELM only, and on OLLB it is not clearly harder than the median. The median is cheap in points, and the minimum is cheap when feasible but needs every weak task to be raised and so is limited by the cap. Borda and the Condorcet requirement are cheaper than the mean in points, but their position relative to the median depends on the dataset and the rank stratum (Appendix Table 5), and on HELM Borda reaches only one rank further than the mean (rank 17 against 16). These orderings come with caveats we state next.

Conditions under which the conclusions fail or are not established. (1) *The cap.* Feasibility is determined by the cap, not by the rule, far from the top (oracle rows of Table 1). The OLLB cap uses gains of published fine-tunes over their declared base models; this is selection-biased toward successes and says nothing about what a determined developer could achieve with a new data source. The HELM cap is a transfer of the OLLB MMLU-PRO gain to 57 subjects and has not been measured on HELM. Headline claims are therefore stated per dataset. (2) *Independence across tasks.* We let each task be raised at no cost to others; targeted training can trade off tasks, which would make the true cost higher. (3) *Fixed rivals.* Other developers do not respond; leaderboards with many strategic developers need a game model such as that of Chen et al. [2026]. (4) *Score tables, not training.* The investment is in score points, not compute, data or money, so "cheap" means few score points, not little effort. (5) *Ranks within noise.* The margin of 10^{-3} points that defines "first" is below the evaluation noise of the scores; a win by that margin would not survive resampling (Appendix Figure 3), and the intervals for the median and minimum are wide. (6) *Rule coverage and size.* Rank rules were solved exactly for the 50 best OLLB targets only, so their feasibility fractions there cannot be compared with those of the score rules; the Condorcet requirement is a sufficient condition for Condorcet-consistent rules and therefore an upper bound for Copeland-type rules; Kemeny and Schulze were not studied; Copeland was solved exactly on HELM only, where it coincides with the Condorcet requirement (Section 5). (7) *Small n and small M.* OLLB has six tasks and HELM 78 targets, so rank bins on HELM hold few targets and their intervals are wide; scaling in n comes from HELM subsets only and is a statement about random subsets of one benchmark family. (8) *Data snapshot.* Both leaderboards are frozen snapshots (OLLB 2025-03, HELM v1.13.0); we do not claim anything about later models. (9) *Not done.* Kemeny and Schulze rules were not solved (the Condorcet requirement bounds them above); exact rank-rule solutions for OLLB cover the 50 best targets and a top-500 slice, not all 4495; and caps for HELM were not measured on HELM. We did not study tasks beyond the six OLLB and 57 HELM scores.

Broader impact. The analysis tells leaderboard designers how exposed each rule is to targeted training under stated assumptions, which supports choosing rules and caps (for instance holding out sub-datasets) deliberately. It could also be read as a recipe for gaming a ranking. We think the risk is small: it needs no new capability beyond fine-tuning on data a developer already can use, and our results indicate that for most models it is infeasible under the observed gains. We do not evaluate any model's training.

7 Reproducibility

All numbers, tables and figures are produced by the supplementary code from the pinned public data (versions, checksums, seeds, commands, expected run times and a table mapping every figure and table to its generating script are in `REPRODUCE.md`). The solvers are deterministic; random draws use seed 0. The full pipeline needs no GPU and needs about 8.4 hours of summed job time on a shared

two-CPU container when the steps are run one after the other (per-step times in REPRODUCE.md; most of it is the Monte-Carlo runs with the exact rank-rule solvers and the dataset-slice variants). Neither leaderboard declares a licence for its score data; we release only derived score tables. No material about the published study that motivated this question was searched for or read, and none was encountered (reproduction notes, section on forbidden material).

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A Proofs of Propositions 1–3

Proposition 1. $f(y) = \frac{1}{n}\sum_t y_t \geq \tau$ iff $\sum_t (y_t - x_t) \geq n(\tau - f(x))$, and $\sum_t (y_t - x_t) \leq \sum_t h_t$ with each $0 \leq y_t - x_t \leq h_t$; so feasibility is $\sum_t h_t \geq n(\tau - f(x))$, i.e. $\sum_t c_t \geq n\tau$, and the minimum of the (linear) total is the requirement itself, attained by any feasible spread. K is minimised

by filling the largest headrooms first, since touching k tasks can supply at most the sum of the k largest h_t. The uniform raise by $\tau - f(x)$ attains L_∞, and no spread has a smaller maximum because the average raise is $\tau - f(x)$.

Proposition 2. $\min_t y_t > \tau$ requires $y_t > \tau$ for every t, hence $c_t > \tau$ wherever $x_t \leq \tau$ (tasks with $x_t > \tau$ need no change). The cheapest y_t on a deficient task is τ (up to ε), so $L_1 = \sum_t (\tau - x_t)^+$, and K counts the deficient tasks. Both are forced.

Proposition 3. For odd n the median exceeds τ iff at least m coordinates exceed τ. Tasks with $x_t > \tau$ already count. We need $m' = (m - \#\{x_t > \tau\})^+$ more tasks. Lifting task t above τ costs at least d_t (and is possible iff $c_t > \tau$); costs are separable, so the cheapest set is the m' smallest finite d_t and $K = m'$, which cannot be smaller because each lifted task adds one to the count.

B Further results

This appendix holds the solver check (Table 3), the robustness of the investments to evaluation noise and task resampling (Figure 3), the results under the oracle cap `ceil` (Figure 4), where every target is feasible and only the rule differs, and the effect of the cap on feasibility and cost on OLLB (Figure 5) and HELM (Figure 6).

Table 3: Closed forms (Propositions 1–3) against the exact solvers and an independent generic LP/MILP (60-target sample), over all caps. Generated by `figures/src/tables.py` from `results/theory_check.csv`.

data	rule	reference	max$\lvert\Delta L_1\rvert$	max$\lvert\Delta K\rvert$	feas. mismatches	caps
OLLB	mean	exact solver	5.7e-14	0	0	7
OLLB	mean	generic LP/MILP	8.5e-14	0	0	7
OLLB	min	exact solver	0.0e+00	0	0	7
OLLB	min	generic LP/MILP	4.3e-14	0	0	7
OLLB	median	exact solver	solver only	–	–	7
HELM	mean	exact solver	4.5e-13	0	0	8
HELM	mean	generic LP/MILP	1.8e-12	0	0	8
HELM	min	exact solver	0.0e+00	0	0	8
HELM	min	generic LP/MILP	1.3e-12	0	0	8
HELM	median	exact solver	5.7e-14	0	0	8
HELM	median	generic LP/MILP	1.5e-12	0	0	8

C Sensitivity to analysis choices

Table 4 repeats the primary analysis under changes of the margin, the data and the cost; Table 5 compares each rule with the mean on targets feasible under both. Medians of K and L_1 are among feasible targets and must be read with the feasible share.

Margin. Requiring the target to lead by 1 or 2 points instead of 10^{-3} lowers the feasible share a little (OLLB mean $0.12 \to 0.11 \to 0.10$; HELM mean $0.19 \to 0.15 \to 0.12$) and raises the cost of targets feasible in both (mean: $\times 1.08$ and $\times 1.17$ on OLLB, $\times 1.33$ and $\times 1.73$ on HELM). The ordering of the rules by feasible share on HELM (median above mean) is unchanged. *Merged models.* Dropping models flagged as merges leaves the OLLB shares and the mean and median costs unchanged (ratio 1.00); the minimum rule becomes somewhat cheaper (0.81) with a similar share. *Raw scores.* On un-normalised scores the oracle-cap costs of the mean and median are 0.85–0.86 of the normalised ones and that of the minimum 1.61 times larger. With a $+10$ cap the median is feasible for almost no target (0.00 against 0.05), so the effect of a tight cap depends on the score scale, and we do not compare caps across scales. *Top-500 slice.* Restricting OLLB to the 500 best models makes first place reachable for 0.95 (mean), 0.79 (median) and 0.40 (minimum) of the targets; Borda and plurality (0.91, 0.56; one Borda instance censored) and the Condorcet requirement (0.30) are exact on this slice. *Task weights.* With random positive weights on the tasks the weighted mean costs less than the unweighted one on targets feasible in both (0.54 and 0.70 of the cost for Dirichlet(1) and Dirichlet(5) weights on OLLB, 0.43 and 0.68 on HELM, 0.38 for a one-hot weight on OLLB with 14 jointly feasible targets): unequal weights let a developer concentrate on a heavily weighted task. *Noise units.* In units of the per-task evaluation-noise standard error the median L_1 among feasible targets is 4.3

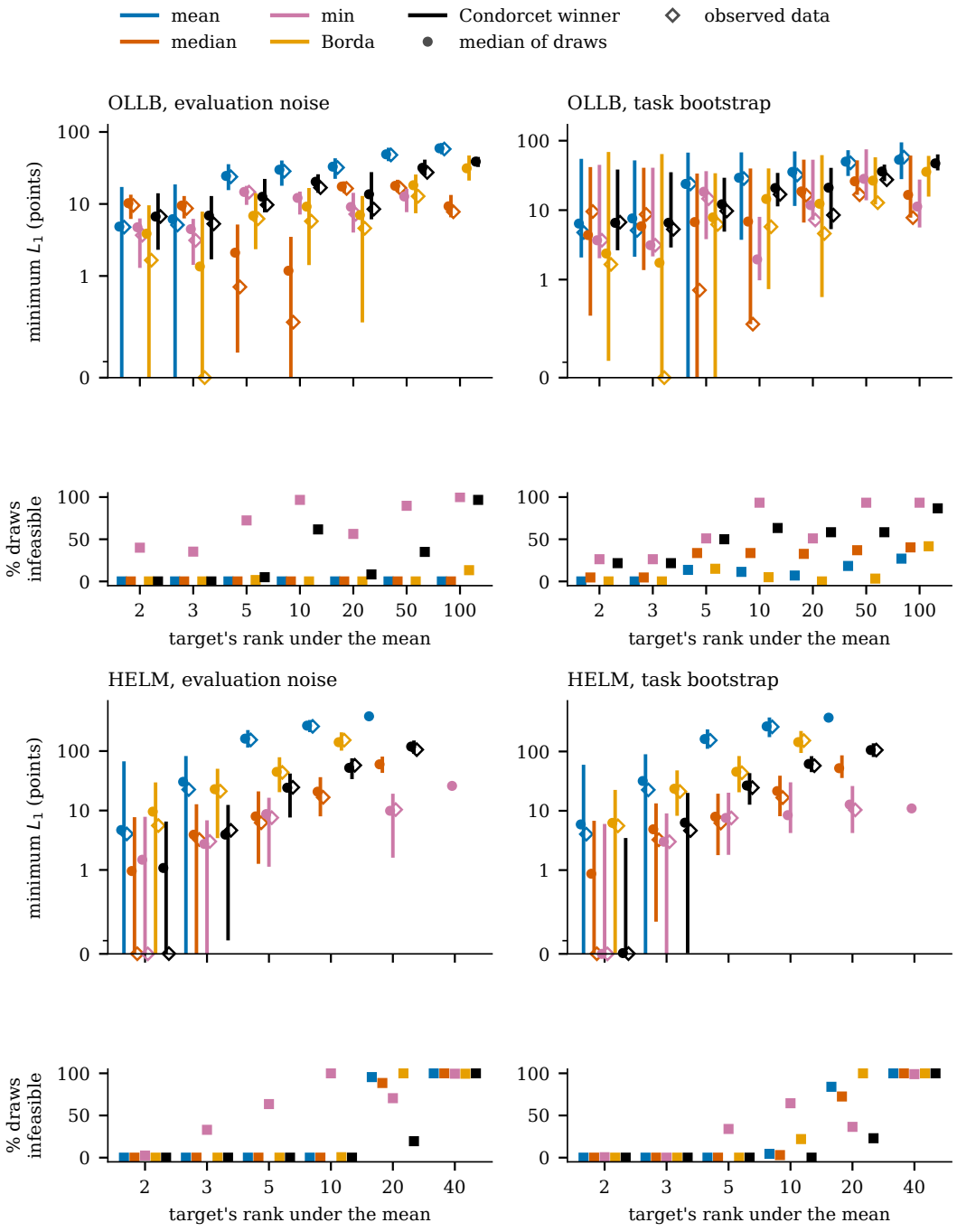


Figure 3: Stability of the minimum total gain L_1 (top) and share of resampled leaderboards in which the target cannot become first (bottom) under evaluation noise (left) and a bootstrap over tasks (right), on OLLB (top; 7 targets, ordinal rules for the 50 best only) and HELM (bottom; 6 targets). Diamonds mark the observed value, dots and bars the median and 95% interval over feasible draws. Generated by `figures/src/fig_robust.py`.

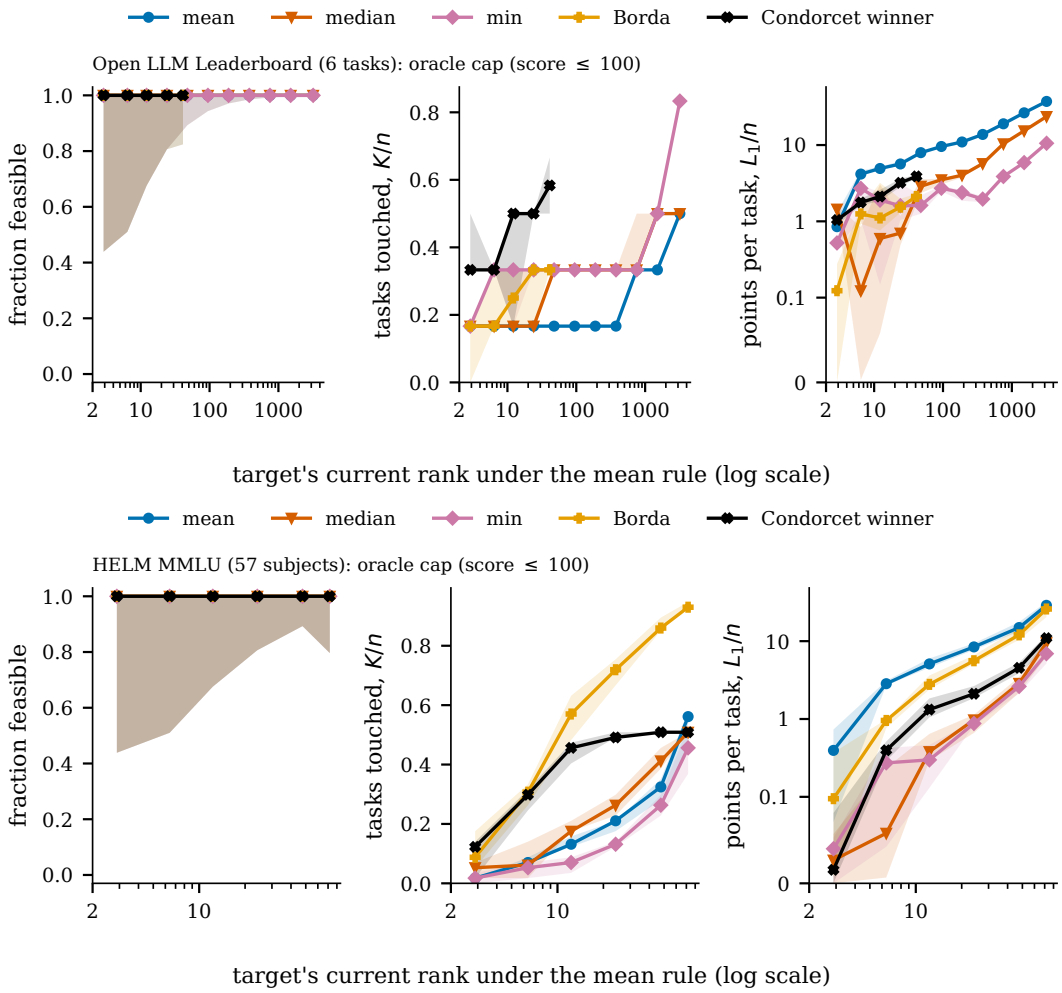


Figure 4: As Figure 1 with the oracle cap `ceil` (scores up to 100): a lower bound on any required investment; every target is feasible, so the differences between rules are not caused by the cap.

(mean) and 1.6 (minimum) on OLLB, and 17.5, 2.5 and 0.6 (mean, median, minimum) on HELM; the minimum rule's investments on HELM are thus below the noise of a single score.

D Hypotheses fixed before the full results

The plan fixed three hypotheses from the closed forms before the full result set was available. H1: with loose caps, score-sum rules need the same L_1 regardless of K, while order-statistic and majority rules need more tasks but possibly fewer points than the mean. H2: required investment grows with the target's initial rank; feasibility falls as caps tighten; rule ordering can change with the cap. H3: for fixed per-task caps the number of tasks that majority and median rules need does not shrink with n, while the mean rule's L_1/n depends on the density of gaps. Anything beyond these is exploratory.

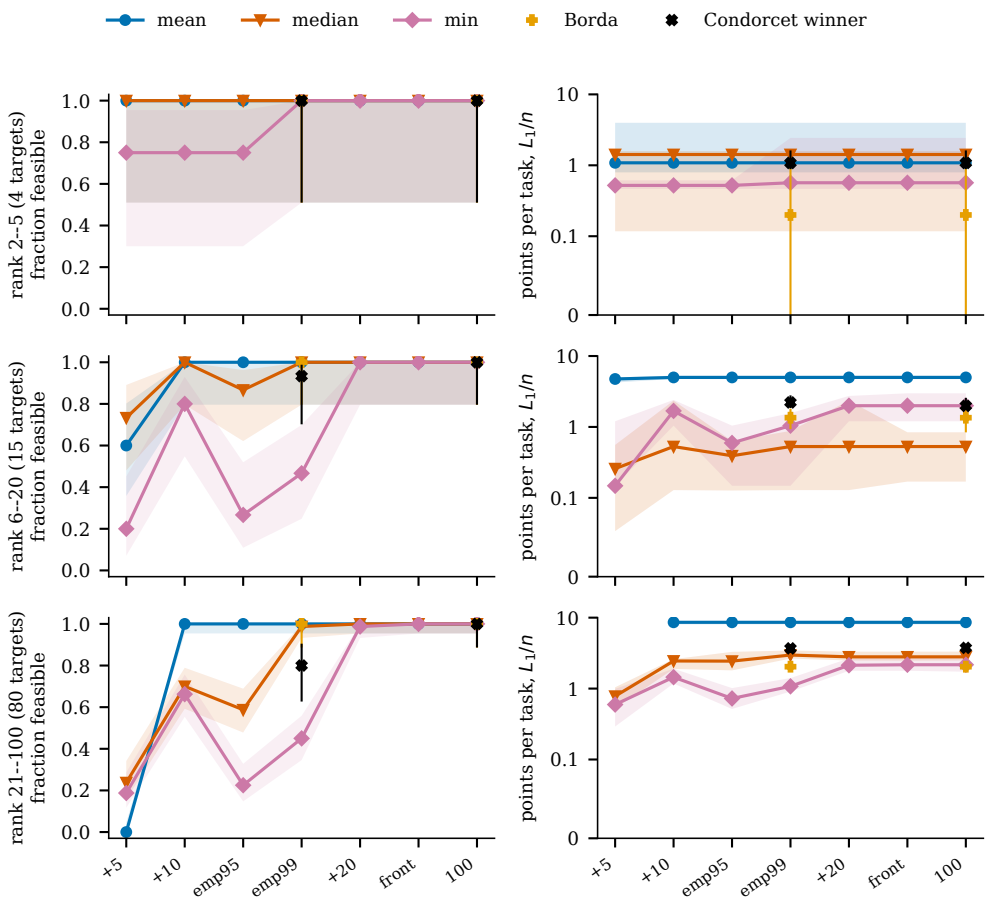


Figure 5: OLLB: effect of the cap on feasibility (left) and cost among feasible targets (right) for three rank strata. Caps run from tight to loose, as on the horizontal axis (+5 points per task, +10, `emp95`, `emp99`, +20, best rival score, 100), ordered by the share of targets feasible under the mean. For the mean rule the cost among feasible targets does not depend on the cap (Proposition 1); only the feasible fraction does. Generated by `figures/src/fig_cap_grid.py`.

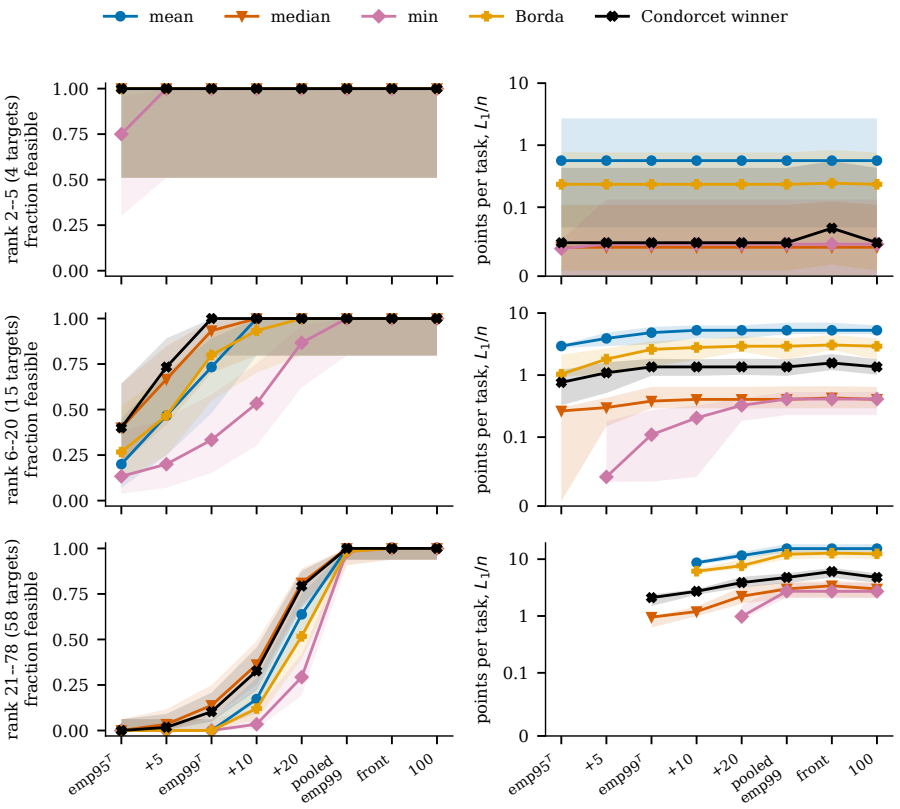


Figure 6: HELM: as Figure 5 for the HELM caps in the order of the horizontal axis (`emp95T`, $+5$ points per subject, `emp99T`, $+10$, $+20$, the pooled `emp99`, the best rival score, 100). The transferred cap `emp99T` lies in the steep part of the curve: for the targets at ranks 21–78 no target is feasible under the mean or the minimum at `emp99T`, 10 of 58 under the mean with $+10$ points and 37 of 58 with $+20$ points (median: 8, 21 and 47). Generated by `figures/src/fig_cap_grid.py`.

Table 4: Sensitivity of the primary result (cap `emp99` on OLLB, `emp99T` on HELM unless stated). n: targets (weight draws for random weights), censored: targets the solver could not decide, feas. share with the baseline share in parentheses, median K and L_1 among feasible targets, ratio of L_1 to the baseline over targets feasible in both (number in parentheses). Generated by `figures/src/tables.py` from `results/summary/summary_variants.csv`.

variant	cap	rule	n	censored	feas. share (base)	median K / L_1	L_1/L_1^{base} (n_{both})
margin 1 pt	`emp99`	mean	4495	0	0.11 (0.12)	4.0 / 77.1	1.08 (500)
	`emp99`	median	4495	0	0.09 (0.10)	3.0 / 29.5	1.15 (403)
	`emp99`	min	4495	0	0.04 (0.06)	2.0 / 9.6	1.25 (191)
margin 2 pt	`emp99`	mean	4495	0	0.10 (0.12)	4.0 / 82.0	1.17 (466)
	`emp99`	median	4495	0	0.08 (0.10)	3.0 / 32.2	1.34 (341)
	`emp99`	min	4495	0	0.03 (0.06)	2.0 / 10.9	1.53 (115)
no merged models	`emp99`	mean	3788	0	0.11 (0.11)	4.0 / 74.1	1.00 (435)
	`emp99`	median	3788	0	0.09 (0.09)	3.0 / 26.6	1.00 (356)
	`emp99`	min	3788	0	0.07 (0.06)	2.0 / 7.4	0.81 (212)
	`ceil`	mean	3788	0	1.00 (1.00)	2.0 / 183.3	1.00 (3788)
	`ceil`	median	3788	0	1.00 (1.00)	3.0 / 108.0	1.00 (3787)
	`ceil`	min	3788	0	1.00 (1.00)	3.0 / 42.4	0.93 (3787)
raw scores	`ceil`	mean	4495	0	1.00 (1.00)	2.0 / 154.1	0.85 (4495)
	`ceil`	median	4495	0	1.00 (1.00)	3.0 / 92.9	0.86 (4494)
	`ceil`	min	4495	0	1.00 (1.00)	5.0 / 74.7	1.61 (4493)
	`gain10`	mean	4495	0	0.07 (0.03)	5.0 / 48.5	0.74 (113)
	`gain10`	median	4495	0	0.00 (0.05)	2.0 / 15.7	5.65 (20)
	`gain10`	min	4495	0	0.05 (0.10)	2.0 / 12.6	1.50 (217)
top-500 slice	`emp99`	mean	499	0	0.95 (0.95)	3.0 / 70.3	1.00 (473)
	`emp99`	median	499	0	0.79 (0.79)	3.0 / 25.5	1.00 (395)
	`emp99`	min	499	0	0.40 (0.40)	2.0 / 7.6	1.00 (200)
	`emp99`	Borda	499	1	0.91	4.0 / 36.5	0.91 (48)
	`emp99`	Condorcet	499	0	0.30	5.0 / 42.2	1.00 (42)
	`emp99`	plurality	499	0	0.56	2.0 / 29.3	0.70 (36)
random weights	`emp99`	Dirichlet(1)	3500	0	0.91 (1.00)	1.0 / 12.2	0.54 (3128)
	`emp99`	Dirichlet(5)	3500	0	0.99 (1.00)	1.0 / 19.1	0.70 (3440)
	`emp99`	one-hot	42	0	0.33 (1.00)	1.0 / 8.5	0.38 (14)
cost in σ units	`emp99`	mean	4495	0	0.12	– / 4.3	–
	`emp99`	min	4495	0	0.06	– / 1.6	–

variant	cap	rule	n	censored	feas. share (base)	median K / L_1	L_1/L_1^{base} (n_{both})
margin 1 pt	`emp99T`	mean	78	0	0.15 (0.19)	33.0 / 229.1	1.33 (12)
	`emp99T`	median	78	0	0.28 (0.33)	11.0 / 30.1	1.55 (22)
	`emp99T`	min	78	0	0.12 (0.12)	3.0 / 4.7	1.50 (8)
	`emp99T`	Borda	78	0	0.19 (0.21)	31.0 / 116.3	1.16 (15)
	`emp99T`	Condorcet	78	0	0.32 (0.32)	27.0 / 88.0	1.14 (25)
	`emp99T`	plurality	78	0	0.53 (0.58)	9.0 / 35.7	1.38 (40)
margin 2 pt	`emp99T`	mean	78	0	0.12 (0.19)	39.0 / 270.3	1.73 (9)
	`emp99T`	median	78	0	0.21 (0.33)	12.5 / 38.8	2.34 (16)
	`emp99T`	min	78	0	0.09 (0.12)	3.0 / 6.7	2.26 (6)
	`emp99T`	Borda	78	0	0.19 (0.21)	31.0 / 117.2	1.17 (15)
	`emp99T`	Condorcet	78	0	0.32 (0.32)	27.0 / 90.2	1.17 (25)
	`emp99T`	plurality	78	0	0.46 (0.58)	9.5 / 43.4	1.95 (35)
random weights	`emp99T`	Dirichlet(1)	3000	0	0.68 (0.67)	6.0 / 33.6	0.43 (1739)
	`emp99T`	Dirichlet(5)	3000	0	0.67 (0.67)	12.0 / 74.9	0.68 (1836)
cost in σ units	`emp99T`	mean	78	0	0.19	– / 17.5	–
	`emp99T`	median	78	0	0.33	– / 2.5	–
	`emp99T`	min	78	0	0.12	– / 0.6	–

Table 5: Median ratio of L_1 of each rule to that of the mean, over targets feasible under both, with 95% bootstrap interval and (targets feasible under both)/(targets feasible under either). OLLB (`emp99`, top), HELM (`emp99T`, bottom); rank strata under the mean; † Borda and Condorcet on OLLB cover the 50 best targets only. The selection of jointly feasible targets is not random, so read each ratio with its counts. Generated by `figures/src/tables.py` from `results/summary/paired_*.csv`.

rule vs mean	all	rank 2-5	rank 6-10	rank 11-30	rank 31-100
median	0.39 [0.38, 0.40] 435/561	1.38 [0.03, 2.00] 4/4	0.02 [0.00, 0.18] 4/5	0.13 [0.09, 0.29] 20/20	0.37 [0.32, 0.40] 69/70
min	0.11 [0.10, 0.11] 237/581	0.61 [0.35, 0.77] 4/4	– 1/5	0.21 [0.10, 0.28] 12/20	0.14 [0.10, 0.18] 29/70
Borda†	0.27 [0.25, 0.30] 48/546	0.26 [0.09, 0.35] 3/4	0.20 [0.18, 0.36] 5/5	0.27 [0.23, 0.37] 20/20	0.27 [0.25, 0.31] 20/70
Condorcet†	0.52 [0.47, 0.57] 42/546	0.91 [0.41, 1.41] 4/4	0.59 [0.40, 0.59] 5/5	0.51 [0.37, 0.56] 16/20	0.49 [0.47, 0.57] 17/70

rule vs mean	all	rank 2-3	rank 4-5	rank 6-10	rank 11-20	rank 21-79
median	0.07 [0.04, 0.10] 15/26	– 2/2	– 2/2	0.06 [0.00, 0.09] 5/5	0.09 [0.07, 0.11] 6/9	– 0/8
min	0.05 [0.02, 0.06] 7/16	– 1/2	– 2/2	– 2/5	– 2/7	– 0/0
Borda	0.51 [0.32, 0.59] 15/16	– 2/2	– 2/2	0.36 [0.32, 0.59] 5/5	0.55 [0.51, 0.64] 6/7	– 0/0
Condorcet	0.21 [0.12, 0.28] 15/25	– 2/2	– 2/2	0.19 [0.09, 0.28] 5/5	0.28 [0.21, 0.31] 6/10	– 0/6

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: Abstract and introduction state the two dataset-specific findings, the cap assumption and their scope; each is backed by Table 2, Table 1 and Figure 1, and Section 6 lists where they fail.

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